

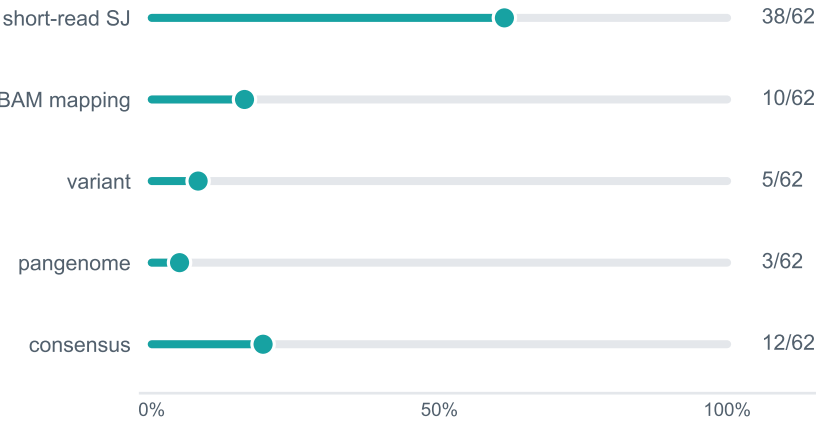
Verdict on 49 novel isoforms

Trustworthy = high/medium-confidence novel · the share you can act on at a glance



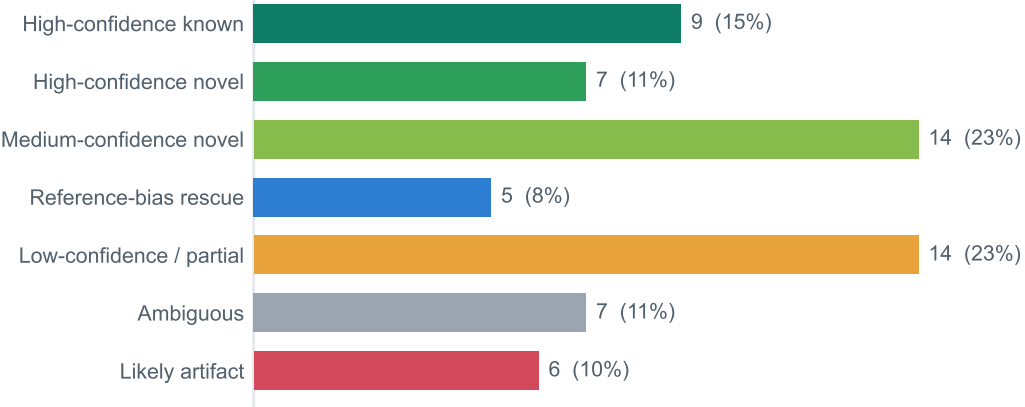
Evidence axes in play

fraction of isoforms for which each axis could be evaluated



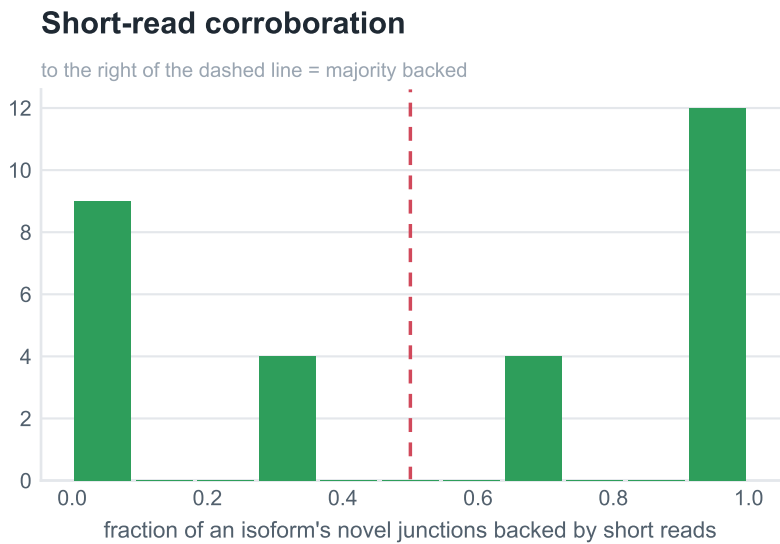
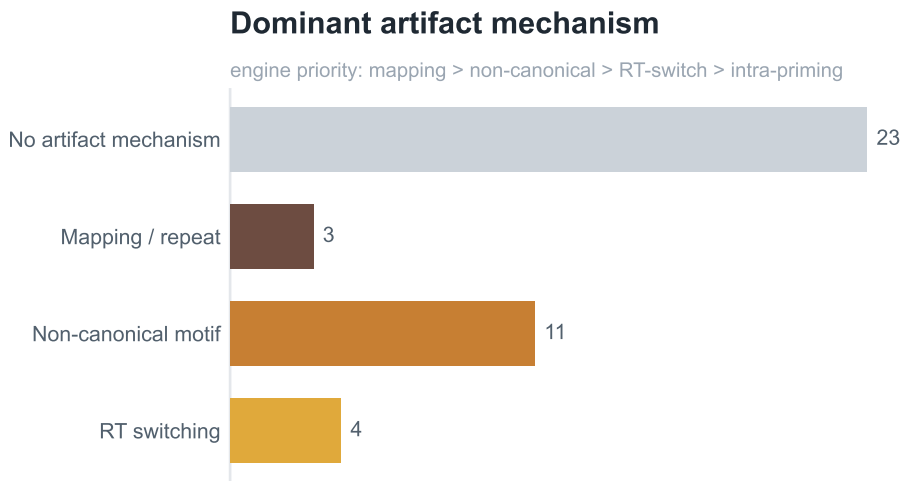
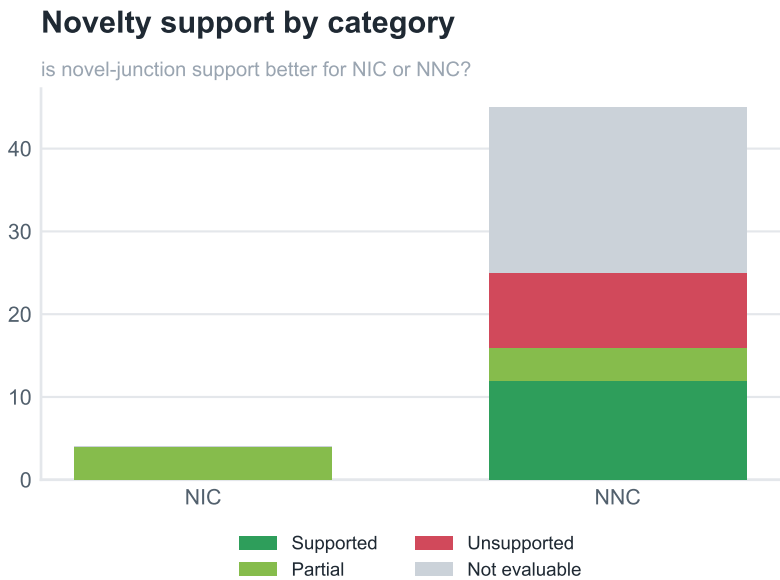
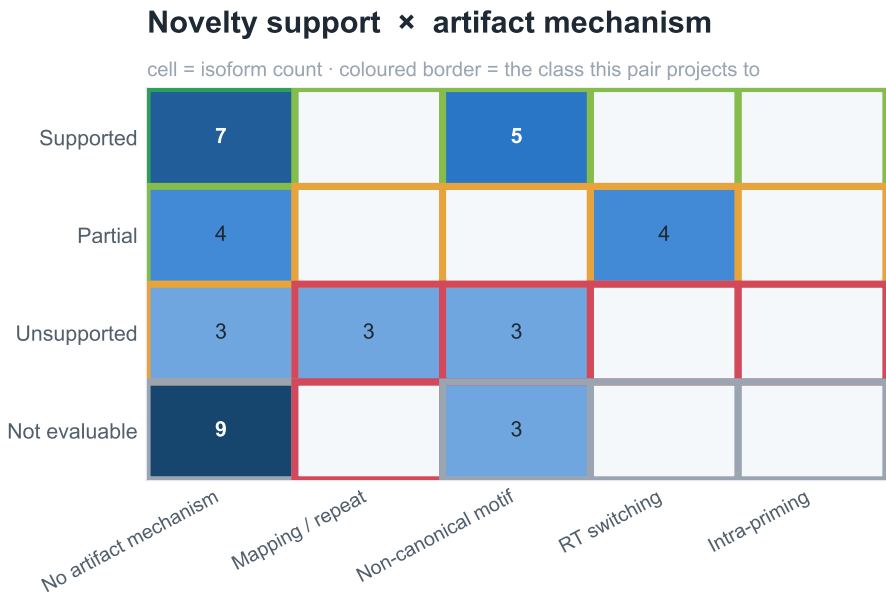
Confidence-class distribution

every isoform · count (share of all)



How each verdict was reached

Every novel isoform is scored on two axes; the pair projects to one confidence class.

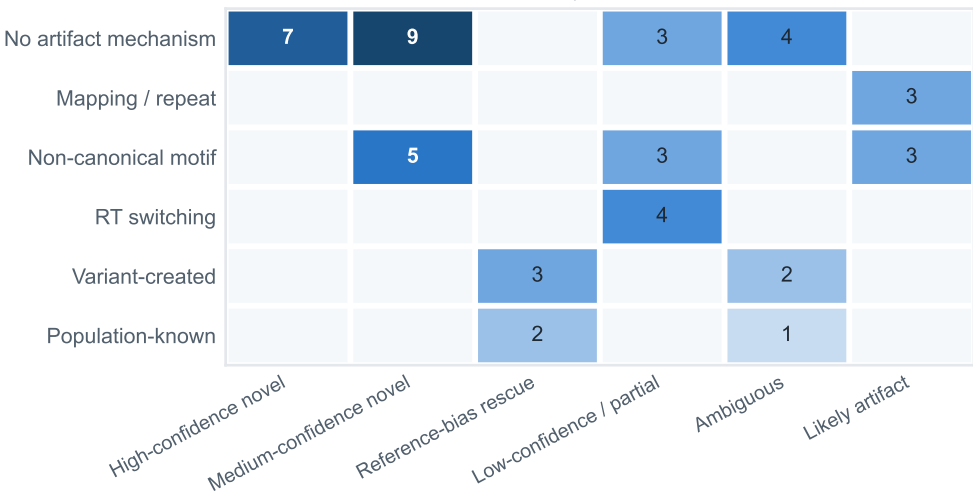


Why calls were down-weighted

Which artifact mechanism drove each verdict, and whether the evidence supports it.

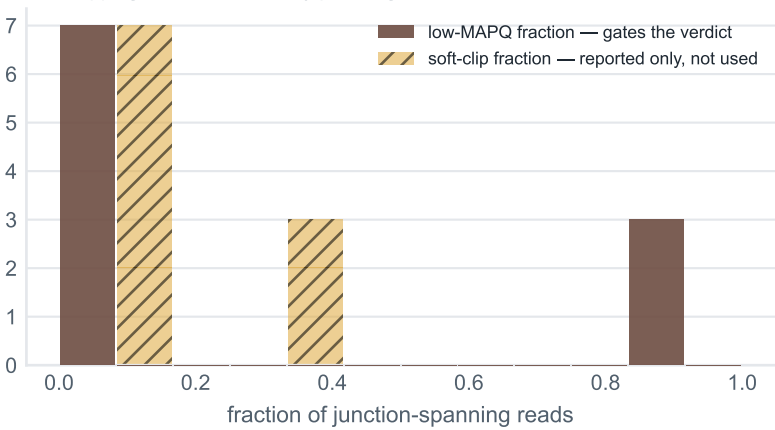
Mechanism → resulting class

where each artifact mechanism ends up



BAM mapping-evidence quality

are mapping artifacts backed by poor alignment?



Structural-signal availability

null is shown as n/a — never silently counted as false

Structural prior	true	false	n/a
Non-canonical motif	11	51	0
RT-switch stage	4	58	0
Caller chain available	62	0	0

Mapping-down-weighted calls (3)

isoform	class	reads	low-MAPQ	suppl.
NNC_map.36	Likely artifact	14	0.86	0.71
NNC_map.37	Likely artifact	14	0.86	0.71
NNC_map.38	Likely artifact	14	0.86	0.71

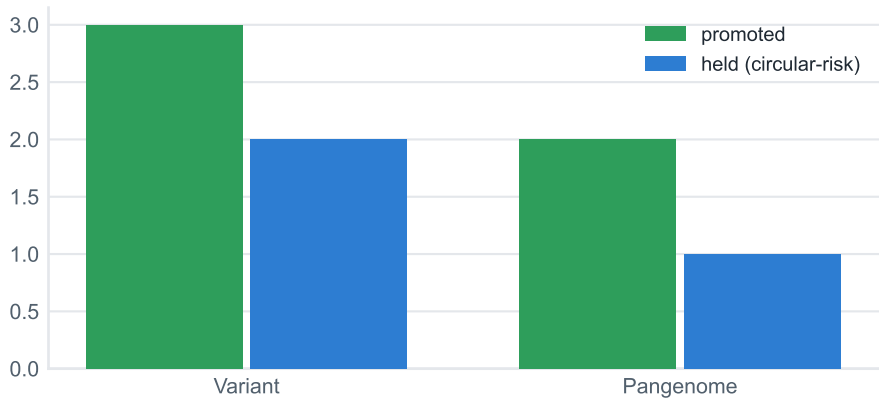
Reference-bias rescue & circularity firewall

A novel junction explained by reference bias is reclassified — but only on independent evidence.

The thesis: nothing is promoted on circular evidence



Rescue axis: promoted vs held

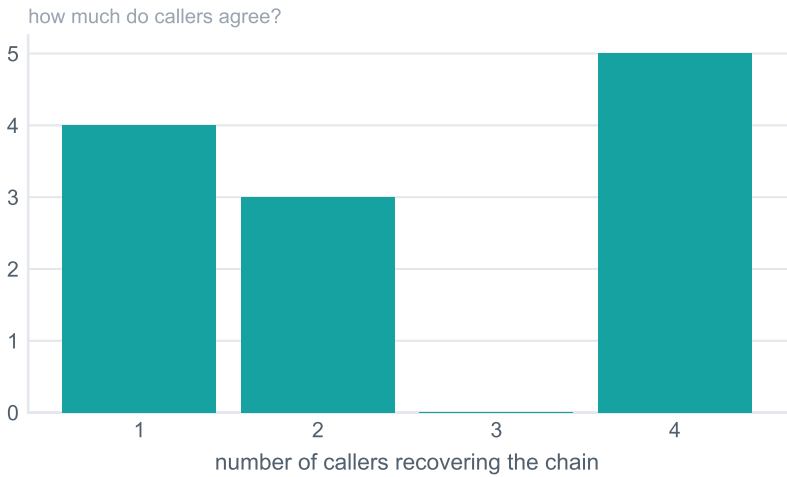


Held-rescue ledger (3)

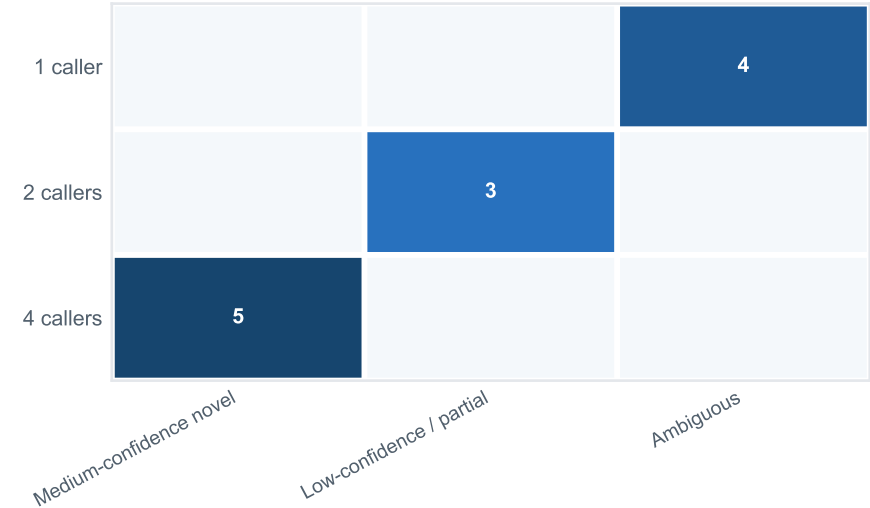
isoform	category	why held (rule trace)
HELD_var.44	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_var.45	not_in_catalog	variant creates canonical motif on haplotype but prove
HELD_pan.46	not_in_catalog	all 2/2 novel junctions realizable on a pangenome path

Caller agreement is methodological corroboration — it promotes to MEDIUM at most, never HIGH.

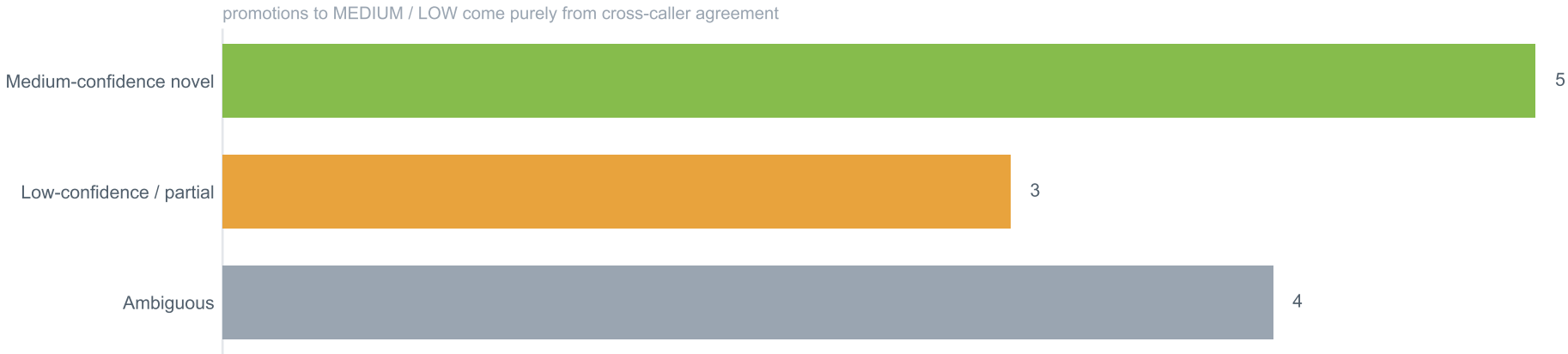
Caller-support distribution



Caller support → confidence class

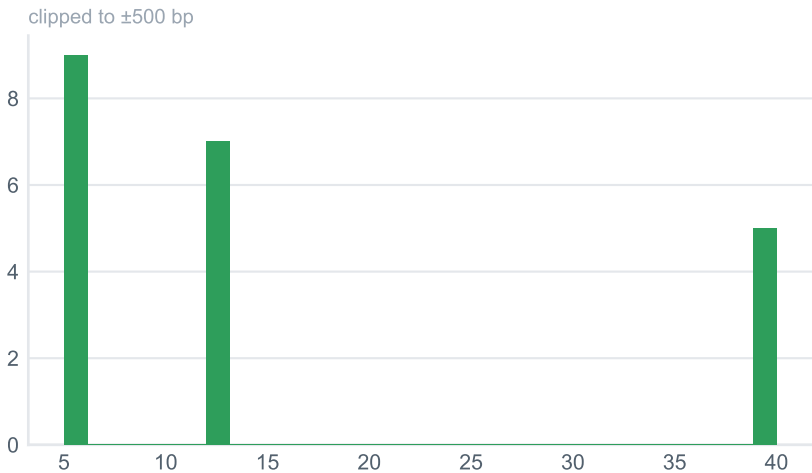


Verdicts of short-read-unknown novels under the consensus axis

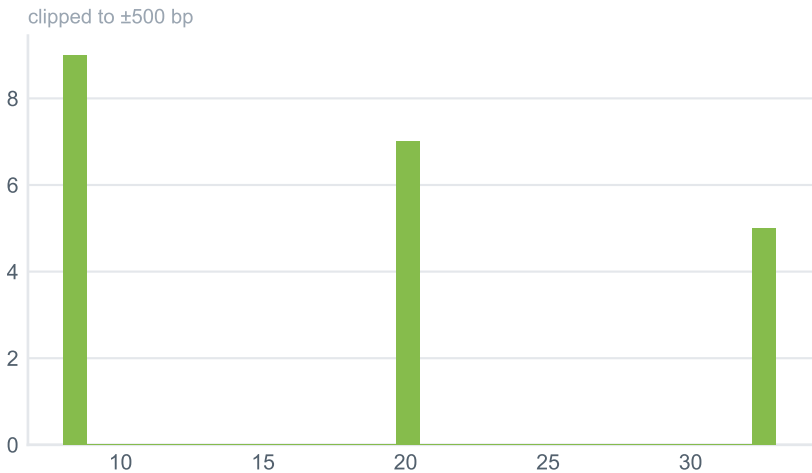


Passed through verbatim — these are NOT used in the PanIsoGuard verdict.

Distance to CAGE peak



Distance to polyA site



Predicted NMD



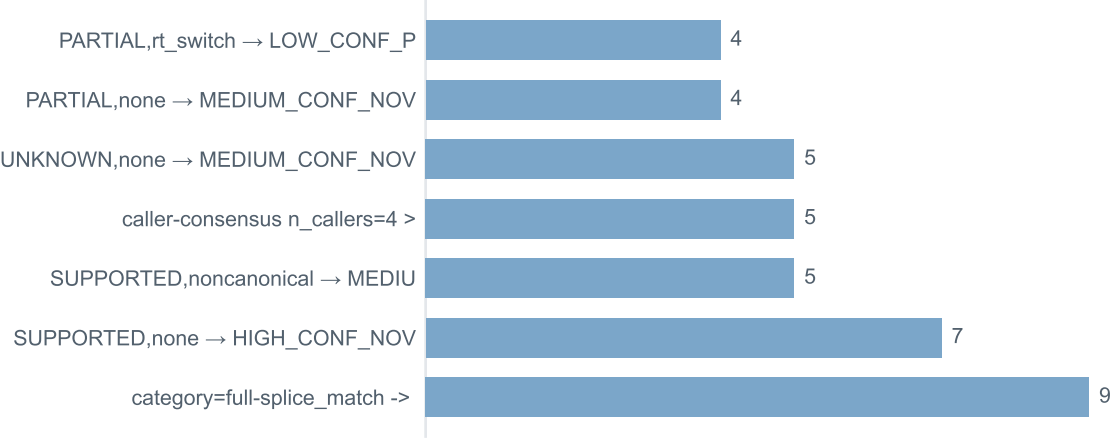
PolyA motif found



Run provenance

Most frequent rule-trace lines

key	value
tool_version	0.0.3
ruleset_version	builtin-0.0.1
sqanti3_version_target	6.0
config	<built-in defaults>
classification	showcase (synthetic demo)
isoforms	showcase (synthetic demo)
ref_gtf	showcase (synthetic demo)
sj_tab	showcase (synthetic demo)
bam	showcase (synthetic demo)
caller_support	showcase (synthetic demo)
axis.short_read	on
axis.catalog	on
axis.bam	on
axis.variant	on
axis.pangenome	on
axis.consensus	on



Defensibility checklist

- ✓ class tally reconciles (provenance ↔ JSONL)
- ✓ firewall held 3 circular-risk rescue(s)
- consensus axis used this run
- axes in play: short-read, BAM, variant, pangenome, consensus
- SQANTI3 priors consumed, never recomputed